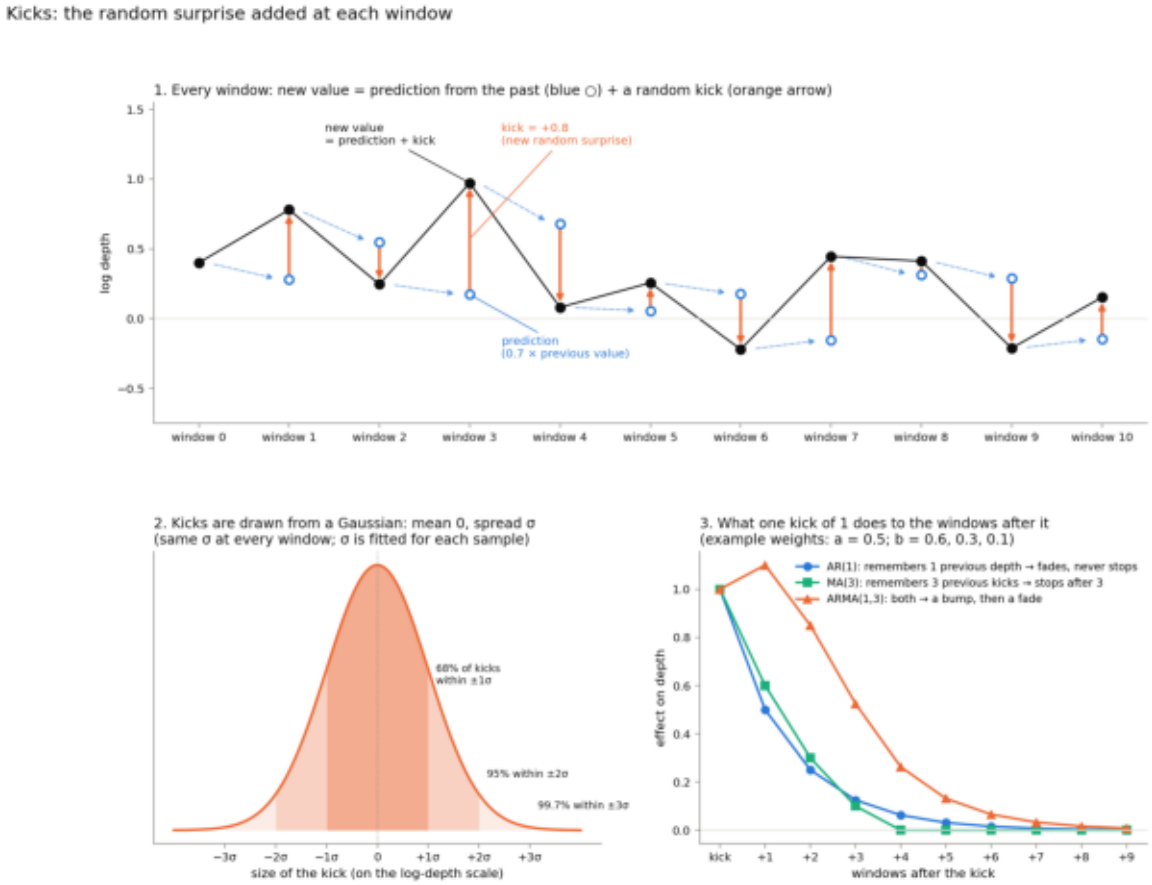
Kicks: the random surprise at each window



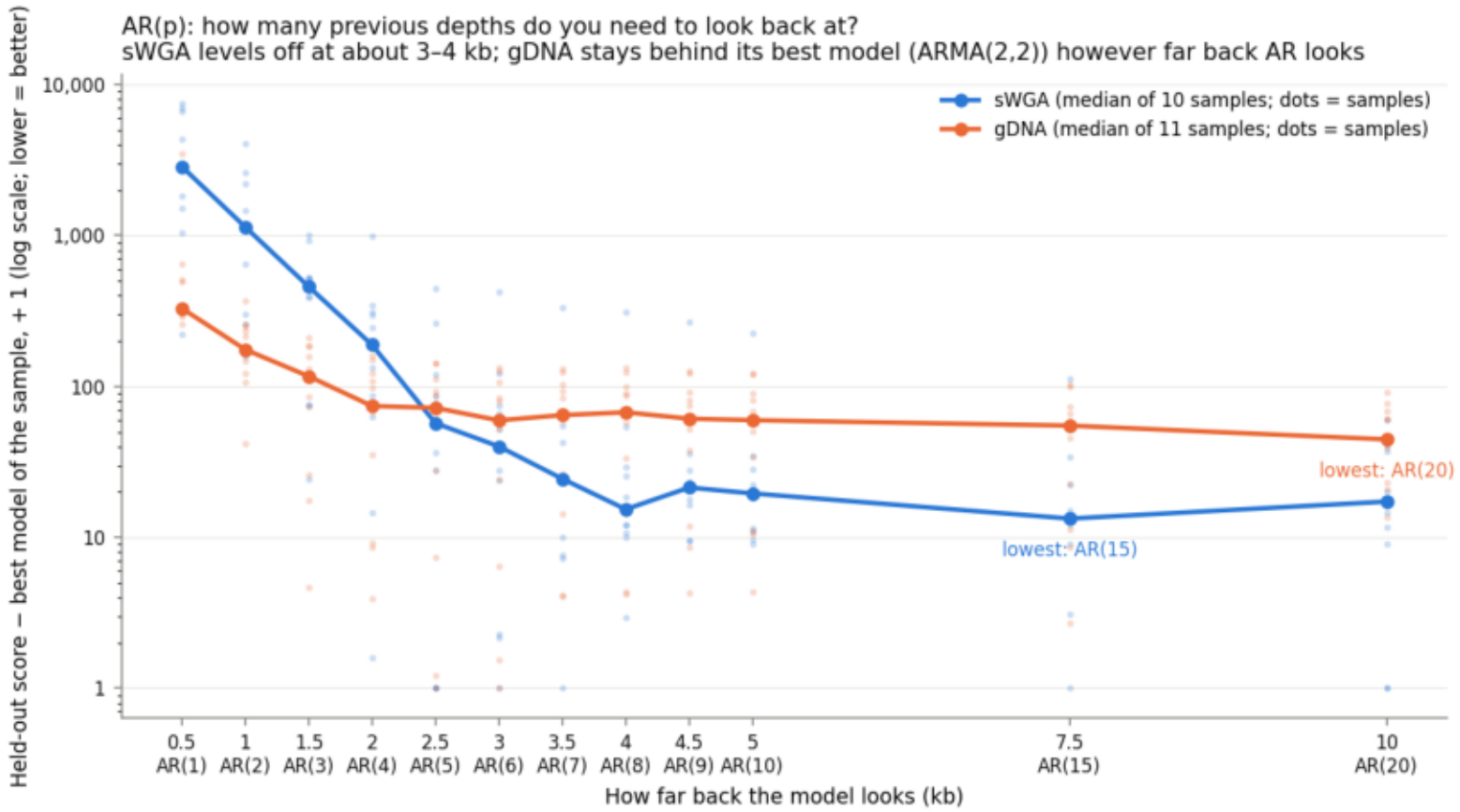
What's on it

- Panel 1: each window's new value = a prediction from the past (blue circle) + a random kick (orange arrow).
- Panel 2: kicks are drawn from a Gaussian with mean 0 and spread σ. 68% fall within ±1σ, 95% within ±2σ, 99.7% within ±3σ.
- Panel 3: what one kick of size 1 does to the windows after it, under three models.

Takeaways

- The kick is new randomness, independent of the past. The prediction is what depends on the past. σ is fitted per sample.
- AR(1) remembers 1 previous depth: a kick fades gradually and never fully stops.
- MA(3) remembers 3 previous kicks: a kick pushes the next 3 windows, then stops.
- ARMA(1,3) does both: a bump for 3 windows, then a fade.

AR: how far back does the model need to look?



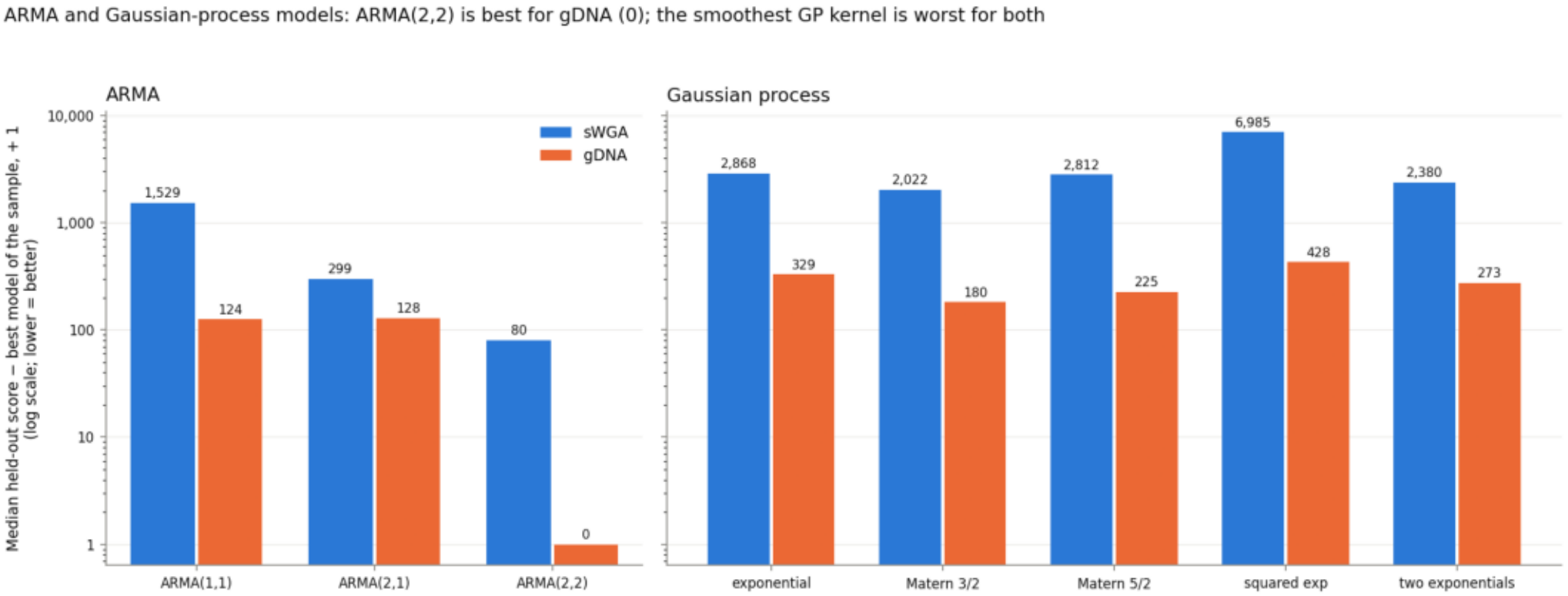
What's on it

- x-axis: how far back the model looks (kb), with the AR order underneath (AR(1) = 0.5 kb, AR(8) = 4 kb).
- y-axis: held-out score, i.e. how far behind each sample's best model (lower = better, log scale).
- Lines = median per sample type; faint dots = individual samples.

Takeaways

- sWGA (blue) falls steeply, from about 2,900 at AR(1) to about 15 at AR(8), then flattens. Looking back about 3-4 kb captures what matters; sWGA depth moves in stretches a few kb long.
- gDNA (orange) flattens at about 45-70 and never reaches 0. No AR does as well as gDNA's best model, ARMA(2,2), however far back it looks.

ARMA and Gaussian-process models



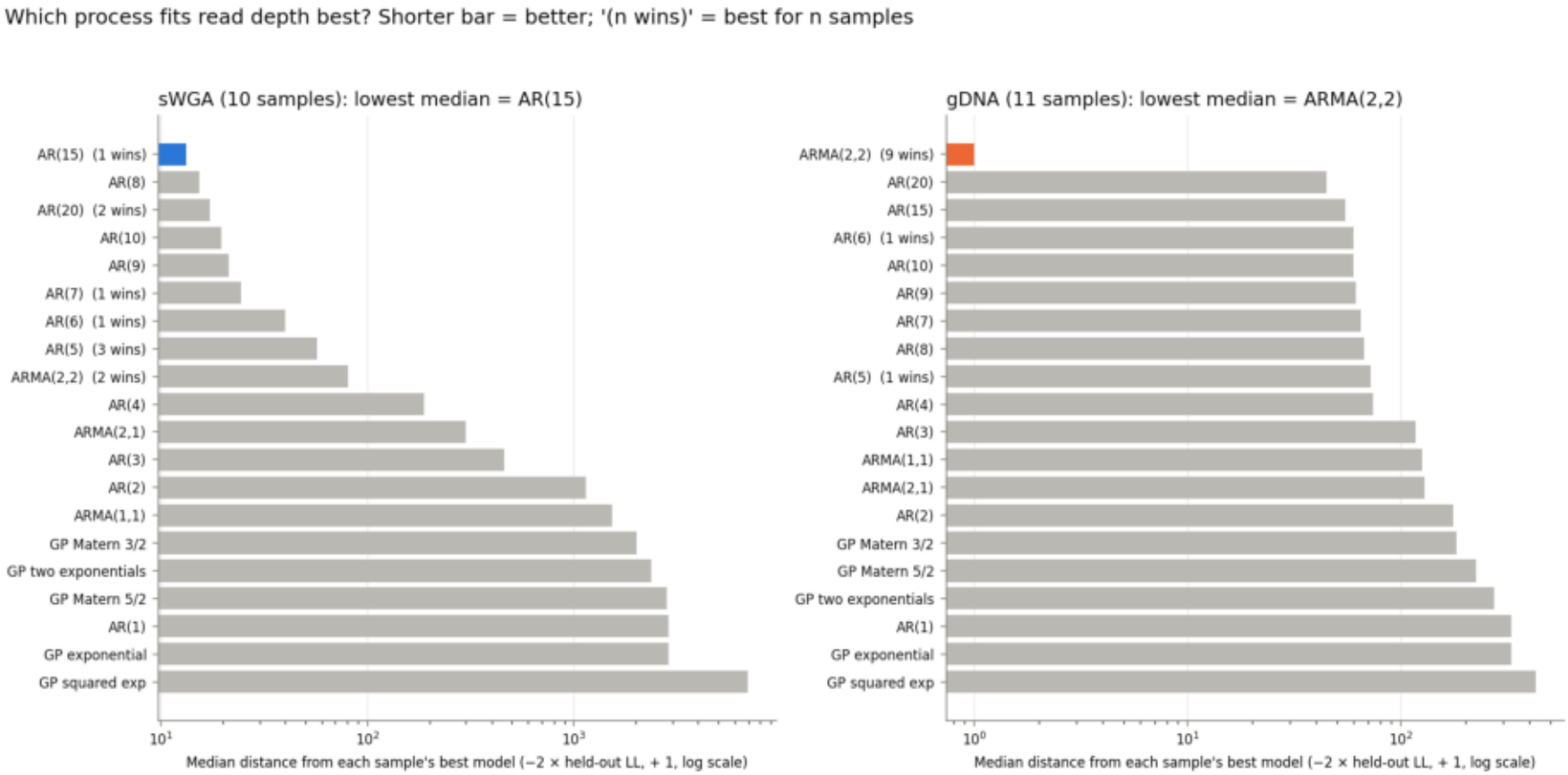
What's on it

- Median held-out score per model, sWGA (blue) and gDNA (orange) side by side, with the number on each bar.
- Left: the three ARMA models. Right: the five GP kernels.

Takeaways

- ARMA, gDNA: ARMA(2,2) scores 0 (best for gDNA); ARMA(1,1) and (2,1) score about 125, so both look-back parts are needed.
- ARMA, sWGA: ARMA(2,2) (80) is behind the mid-length ARs (about 15).
- GP: all far behind. Squared exponential, the smoothest kernel, is the worst for both types: depth is rough, not smooth.
- Check: GP exponential (2,868) matches AR(1) (2,867), because they are the same model.

Every model in order, per sample type



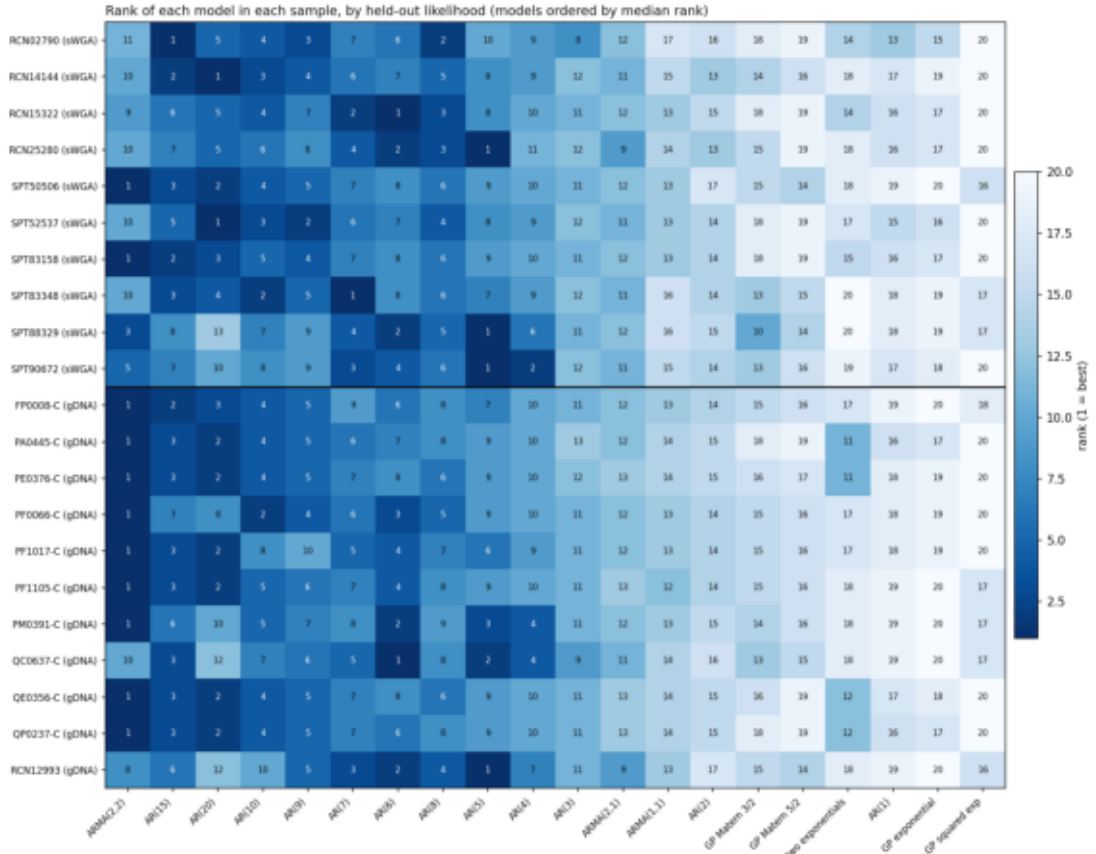
What's on it

- Models sorted from worst (bottom) to best (top) by median held-out score. Shorter bar = better.
- The coloured bar is the best model; '(n wins)' = best for n samples.

Takeaways

- gDNA: ARMA(2,2) is far ahead and wins 9 of 11 samples.
- sWGA: the top is a near-tie of AR(7)-AR(20); the wins are spread across several models, so there is no single sWGA winner.
- Both: AR(1), GP exponential and GP squared exponential are at the bottom.

Which model wins for each sample?



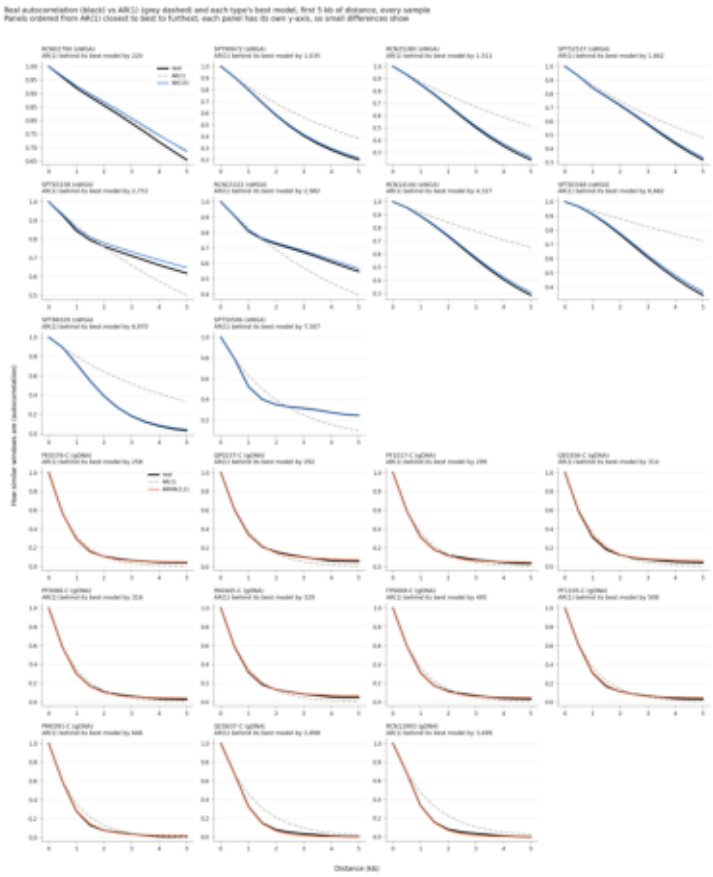
What's on it

- Rows = samples (sWGA above the black line, gDNA below). Columns = models.
- Number = that model's rank for that sample (1 = best, 20 = worst); darker = better.

Takeaways

- gDNA: a solid dark first column, with ARMA(2,2) ranked 1 in 9 samples.
- The exceptions are the two TRAC2 samples (QC0637-C, RCN12993), whose best models are AR(5)-AR(6), like sWGA.
- sWGA: the dark cells are scattered across AR(5)-AR(20), with a different best for different samples.
- The right-hand columns (smooth GPs, AR(1)) are pale for every sample.

Every sample: real vs AR(1) vs the best model



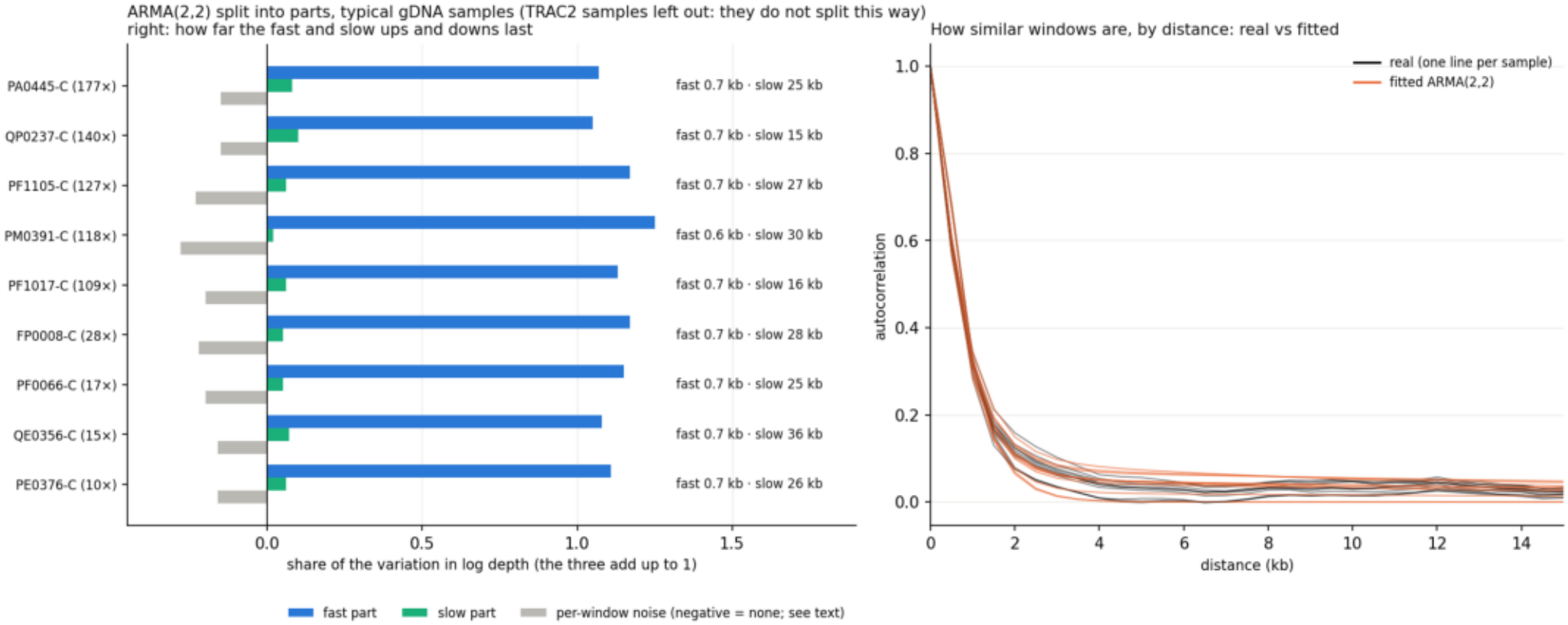
What's on it

- One panel per sample (sWGA on top, gDNA below), ordered from AR(1) closest to the best model to furthest.
- Lines: how similar windows are by distance over the first 5 kb: real (black), AR(1) (grey dashed), the type's best model (colour).
- Each panel has its own y-axis, so small differences show. The title gives how far AR(1) is behind the sample's best model.

Takeaways

- sWGA: in most samples AR(1) misses badly. The real curve bends (a different rate of fading at short and longer distances), which a single exponential cannot follow; AR(15) tracks it almost exactly. Only RCN02790 is close for AR(1).
- gDNA: ARMA(2,2) follows the real curve in every sample. AR(1) looks close by eye in typical gDNA (gaps 258-646), but is clearly off for the two TRAC2 samples (2,898 and 3,499).

gDNA in detail: what ARMA(2,2) says



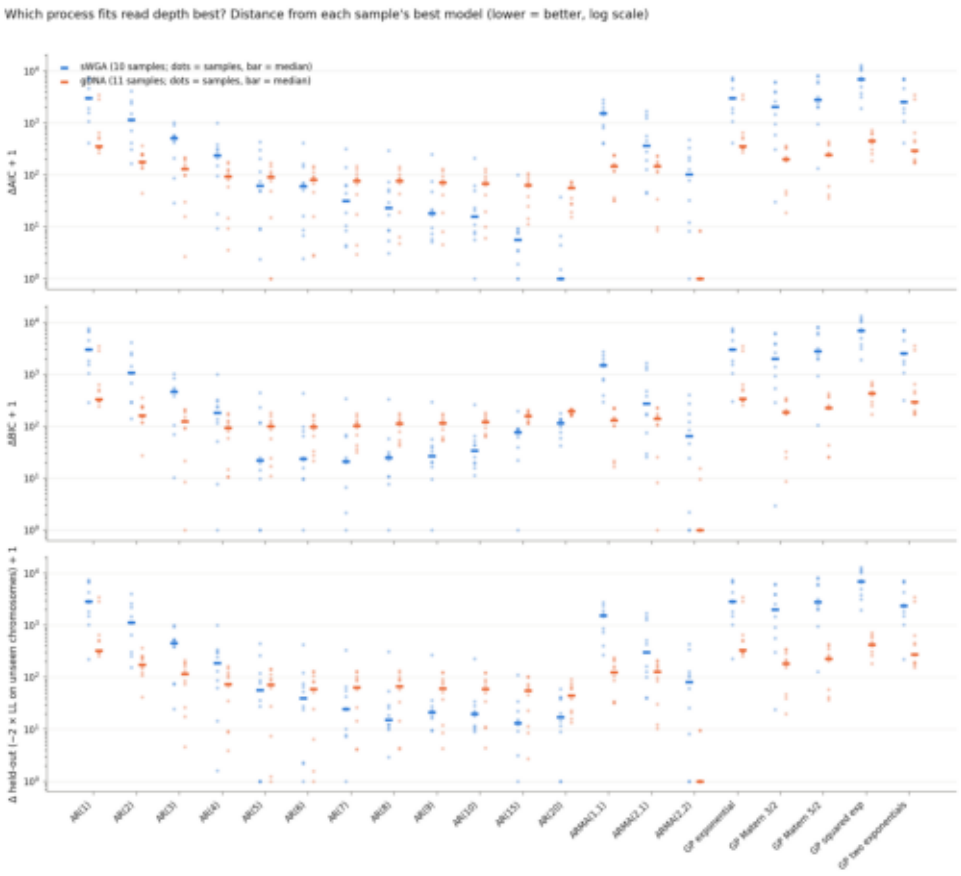
What's on it

- Left: the 9 typical gDNA samples (TRAC2 left out), sorted by depth. ARMA(2,2) split into a fast part (blue), a slow part (green) and a 'noise' part (grey, negative). Text: how long the fast and slow ups and downs last.
- Right: real similarity-by-distance curves (black) against the fitted ARMA(2,2) (orange).

Takeaways

- Almost all the variation is the fast part, lasting about 0.7 kb, in every sample.
- The slow part is small (about 5%) and lasts 15-35 kb.
- The 'noise' share is negative (about -20%): there is no independent per-window noise. Instead, neighbouring windows are extra alike (probably reads straddling window boundaries).
- The same shape from 10x to 177x: it does not depend on sequencing depth. The fit (orange) follows the real curves (black).

All scores: AIC, BIC and held-out



What's on it

- Every model's score in every sample (dots) and the median (bar), sWGA and gDNA side by side, under AIC, BIC and held-out.

Takeaways

- BIC and held-out mostly agree. AIC's penalty is too small for this much data: for sWGA it often picks the biggest model, AR(20).
- Held-out is the main score; BIC is the backup.